

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: Faculty of Engineering and Architecture**
2. **Subject Name: Analysis and Design of Geotechnical Systems L-T-P: 3-0-0 Credits: 3**
3. **Pre-requisites: Geotechnical Engineering (New)**
4. **Syllabus and reference books:**

Syllabus:

Introduction, examples of foundation problems, loading types (drained/undrained, compression/moment/uplift), types of foundation (shallow/deep, isolated, combined, mat, pile, well).

Site characterization, site investigation methods, in situ tests.

Shallow Foundations, contact pressure distribution, shear failure (general/local/punching), bearing capacity, corrections for size, shape, depth, surface slope and compressibility, design of combined footings and rigid mat.

Types of piles, construction methods and their impact on pile behavior, ultimate capacity (static methods for estimating capacities in axial compression, tension and lateral loads), pile load testing (static, dynamic and statnamic), group efficiency (in axial compression and lateral loading).

Settlement of foundations-consolidation (primary, secondary, total and differential settlements); settlement of pile groups.

Stability analysis of earth retaining structures (gravity, cantilever).

Stability of slopes, factor of safety, infinite slope, ordinary method of slices and Bishop's modified method of slices.

Reference Books:

- 1) Bowles, JE. 2017. Foundation Analysis and Design. 5th Edition, Mc-Graw-Hill, New York.
- 2) Das, BM. 2017. Principles of Foundation Engineering. 8th Edition, MindTap. (Paperback)
- 3) Knappett J. and Craig RF. 2019. Craig's Soil Mechanics. 9th Edition, CRC Press.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction, examples of foundation problems, loading types (drained/undrained, compression/moment/uplift), types of foundation (shallow/deep, isolated, combined, mat, pile, well).	2
2.	Site characterization, site investigation methods, in situ tests.	3
3.	Shallow Foundations, contact pressure distribution, shear failure (general/local/punching), bearing capacity, corrections for size, shape, depth, surface slope and compressibility, design of combined footings and rigid mat.	12
4.	Types of piles, construction methods and their impact on pile behavior, ultimate capacity (static methods for estimating capacities in axial compression, tension and lateral loads), pile load testing (static, dynamic and statnamic), group efficiency (in axial compression and lateral loading).	10
5.	Settlement of foundations-consolidation (primary, secondary, total and differential settlements); settlement of pile groups.	9
6.	Stability analysis of earth retaining structures (gravity, cantilever).	3
7.	Stability of slopes, factor of safety, infinite slope, ordinary method of slices and Bishop's modified method of slices.	3
Total number of hours		42